TWISTER

Tornado evolution **WI**th **S**eason and **TER**rain.

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Motivation

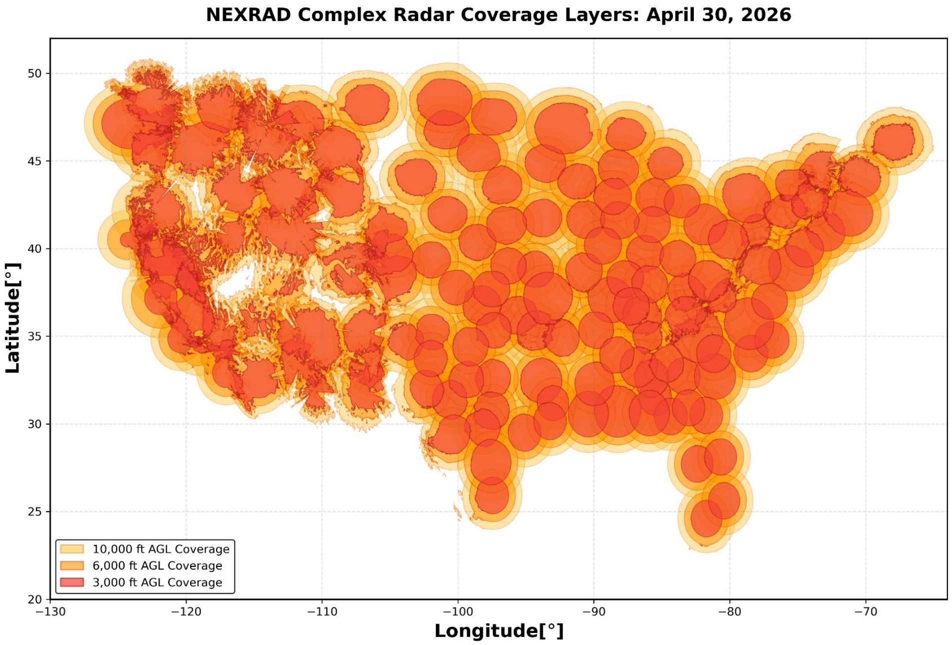
Tornadoes are **sudden, destructive** weather events.

Radar coverage suffers from large gaps.

Motivation: Explore trends in storm event data to be used in increasing tornado safety.

Aim: Identify whether there are underserved regions in radar coverage.

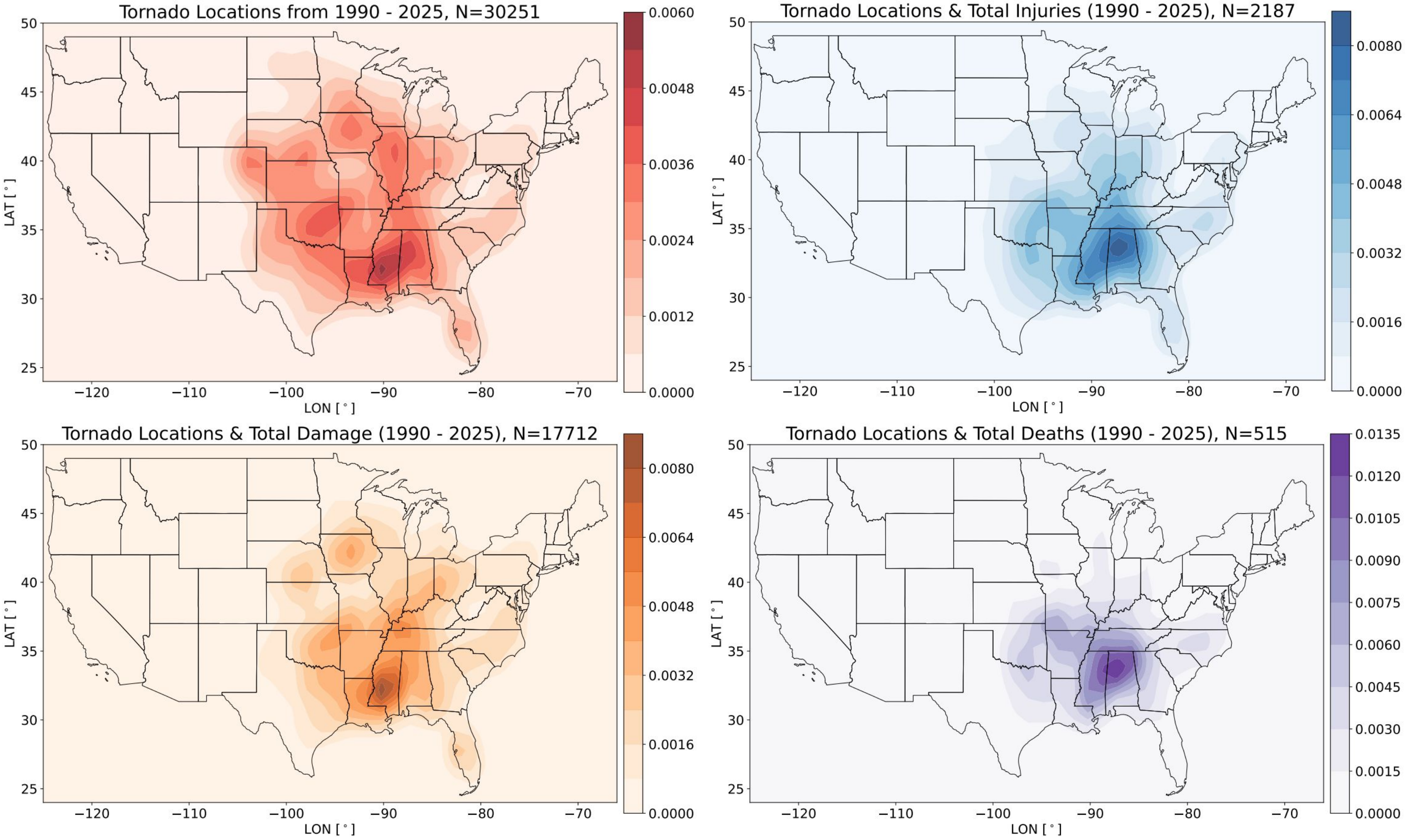
Aim: Determine if tornado alley has shifted in time and location.



NOAA, NEXRAD Radar Locations and Coverage. Apr. 30, 2026.

KDE Maps Show Human Impact

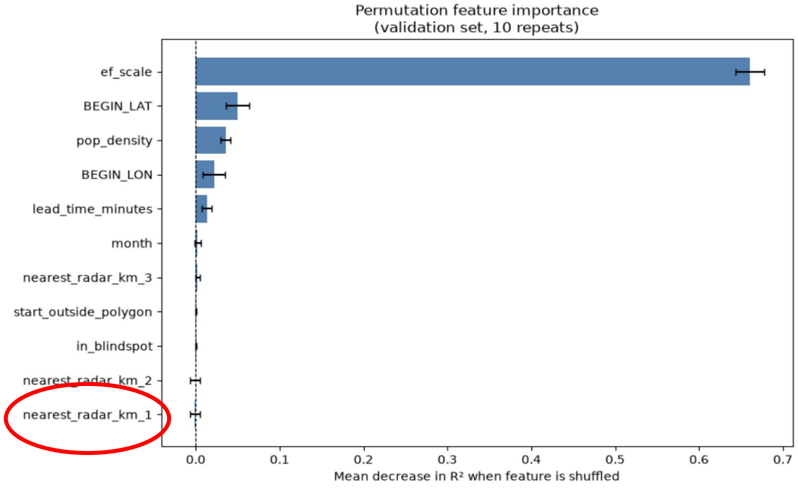
NOAA Storm Events Dataset, 2026.



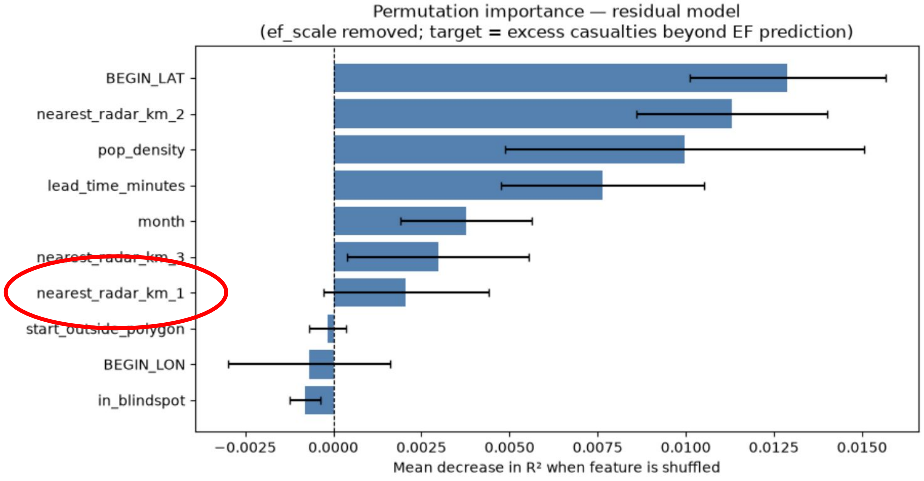
Radar Correlation with Human Impact using Decision Tree Regressors

Hypothesis - Tornado events occurring in radar blind spots are more likely to result in human casualties.

Best predictor of casualties is EF scale (obvious result).



When we marginalize out EF scale, geography becomes the best predictor.

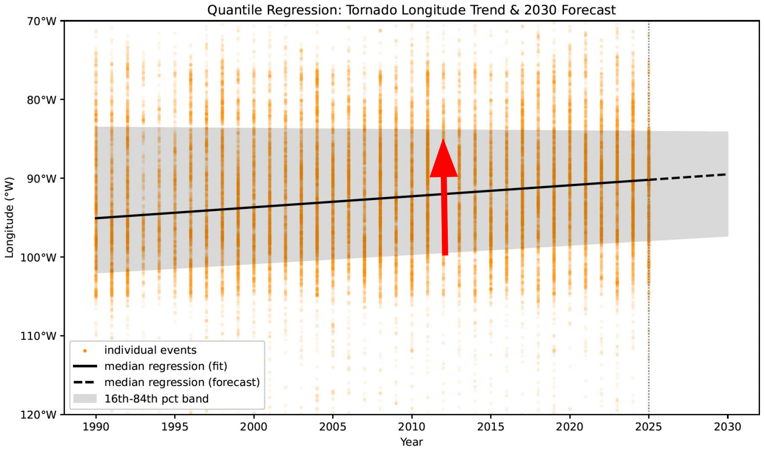


We found no correlation between radar properties and human impact.

Tornado Alley over Years

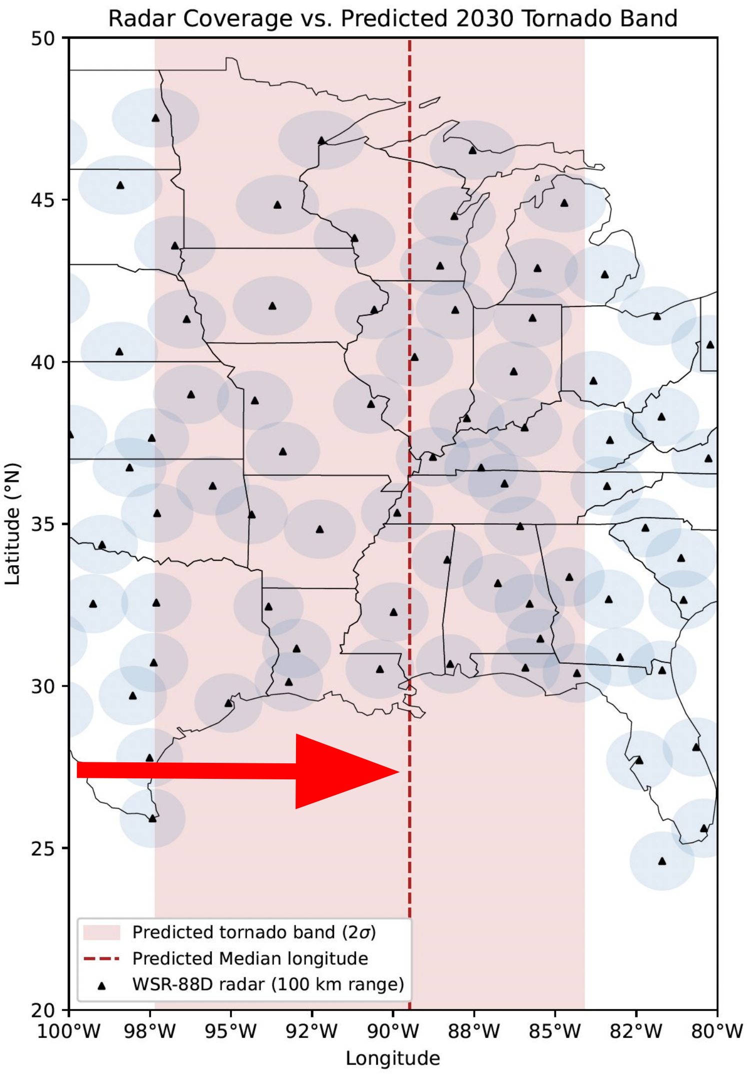
Model geographical shift using **quantile regression** to find the future **region of interest**.

This model predicts tornado alley will be centered over Alabama and Illinois in the next 5 years.



All tornado events **(N=48324)**, **80/20%** for train/test:

1. Stratify by year.
2. **Quantile regression** model with **pinball loss** function.
3. **5-fold CV** check.



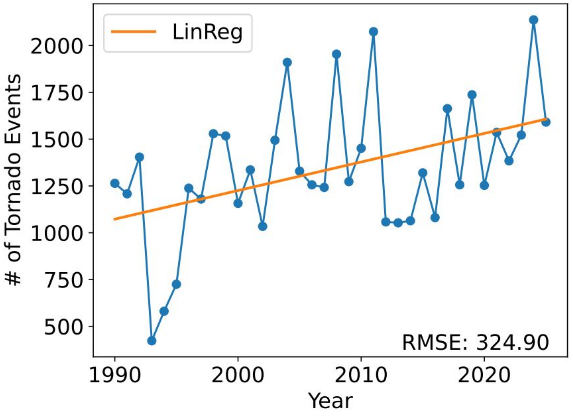
Tornado Seasons over Years

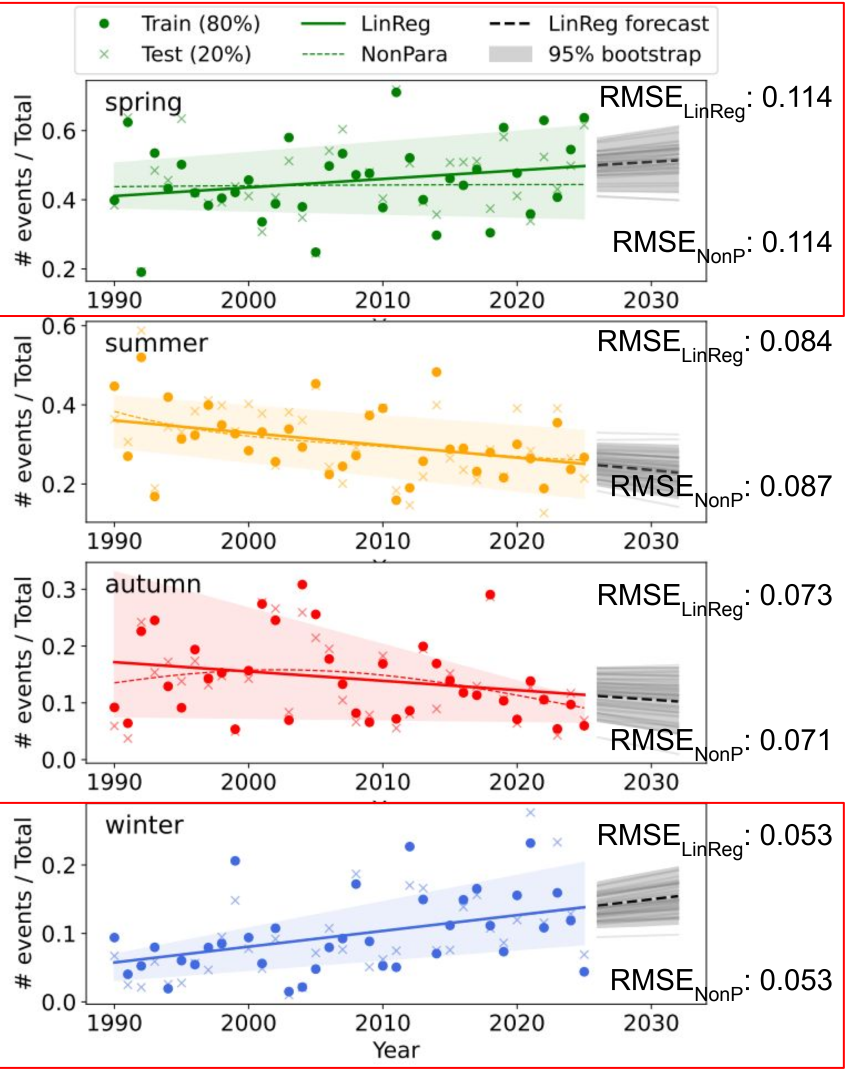
All tornado events **(N=48324)**, **80/20%** for train/test:

1. Stratify by year & use ***fraction***: # events/total
2. Compare **LinReg** & **KernelRidge (NonP)** models
   - KernelRidge - GridSearch for **Alpha** and **Gamma**
   - Cross-Validate: “LeaveOneOut” – **mean RMSE**
3. Use **bootstrapping** (5000 iterations) with LinReg and **extrapolate** fraction to **Year 2030** (**95%** confidence interval)

RMSE’s are similar between linear & NonP

Spring and Winter exhibit positive slopes (shift in tornado season)





Conclusions

The inclusion of new weather radars in current radar gaps **does not obviously decrease human casualties** to tornados.

Instead efforts should be focused on **infrastructure** for access to warning systems and storm shelters and **education** on proper precautions to take when warnings are received.

Tornado alley is expect to evolve in both **season** and **geography**.

The concentration of tornado events is **shifting eastwards**.

Events are also **shifting to the winter and spring** months.

Next steps: Incorporate more data tied to infrastructure (e.g., tornado warning time, hazard mitigation funds per state); convert numerical data to booleans and use classifier algorithm (e.g., random forest classifier) instead; forecast into further timesteps and the implementation of risk assessment for counties within predicted tornado band